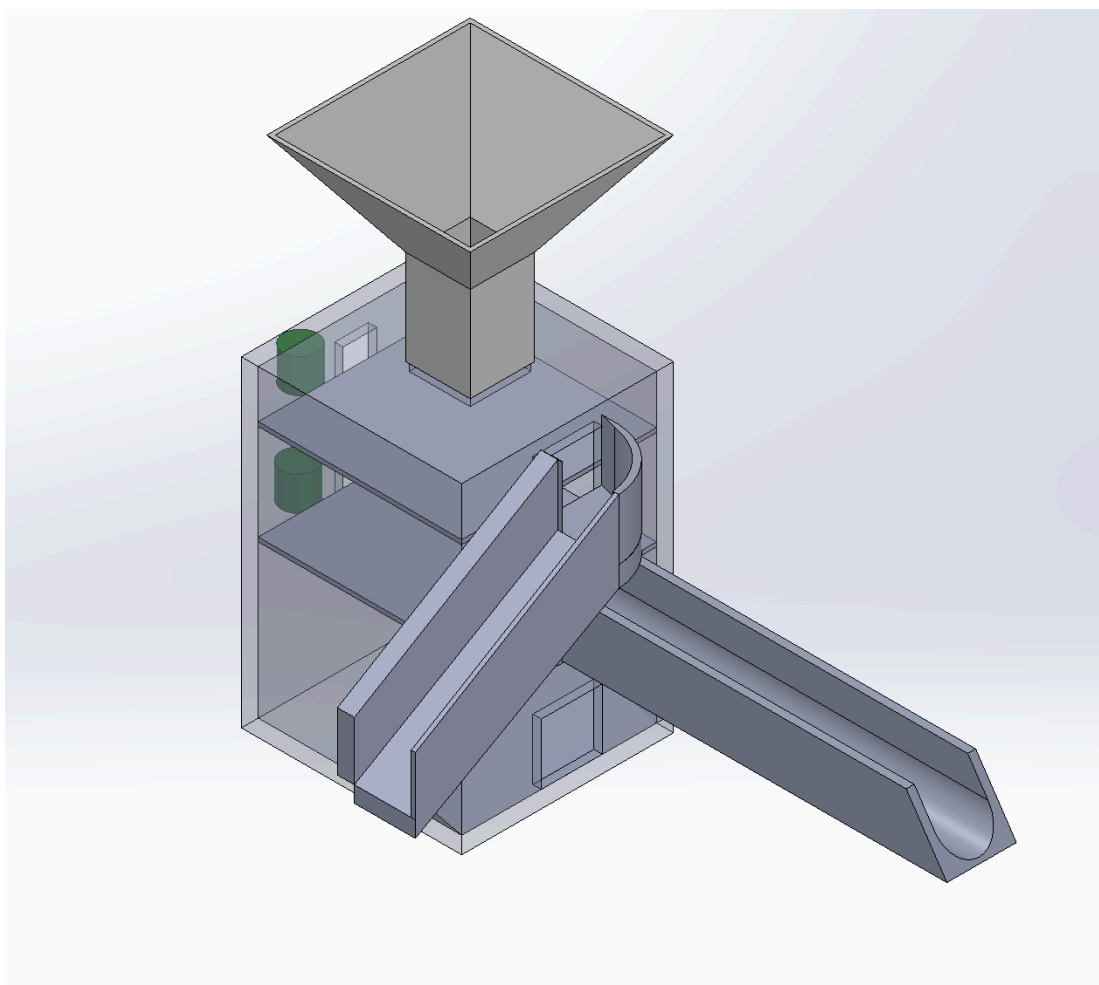


11/07/23
Project 2: Technical Memo
ME 250, Section 3, Fall 2023
Team 10: Appleseeds

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1. Executive Summary:

This technical memo has all relevant information relating to the design of the desired device, as well as the process that the team went through to design said device. To summarize the aspects of this project and report, this item (*item 1*) will discuss the remainder of this document and its components.

After introducing the purpose and intricacies of this project, this report continues with details pertaining to the *problem definition*, *functional analysis*, *concept generation*, *proof of concept*, and *engineering drawings*. The *problem definition* allows the team to determine what the device to be designed is meant to do or what problem it's meant to fix, and this allows the team to set goals and requirements for the project. The *problem definition* is followed by a *functional analysis*, which is when the team decides what specific parts of the device (which is not yet designed) do rather than just looking at the overall function of the whole device. It gets broken down into smaller functions. After this analysis, the design process can truly begin with *concept generation*. This is when the team begins to truly finalize what is expected of the device, and how this will be actualized. This step is then finalized with the *proof of concept* process. This is where the top three-five designs are discussed, which furthers the design process. *Engineering drawings* then allow the team to visualize the top designs, and after this part, creating the device becomes possible.

The design process helped the team recognize the complexity and amount of detail that go into simply crafting a logical and effective design for a device prior to building it. From this process, the team concluded that the design process is an essential practice in developing solutions that can solve problems effectively and efficiently. The design process helped the team come to a better understanding of the finer details within their design, in turn leading to a better and more effective design.

2. Introduction:

Project 2 is a course project for ME250, based around creating a device design through the use of the design process. The objective of this project is to design a device that can continuously and accurately sort, count, and eject ping-pong balls based on their color. The purpose of this technical memorandum is to explain and justify all of Team Appleseed's design choices and decisions throughout the design process.

The said design process took place over the course of six weeks and proceeded through numerous well-defined steps. The first step (which is briefly covered in the paragraph below) was doing background research and performing patent searches while considering ethical aspects. After this, a problem statement was defined, and both objectives and constraints were defined. Using these objectives and constraints, the team created an objectives tree to map out and justify their choices. This is all given under item 3 in the report. Item 4 moves into the functional analysis section of the design process. This breaks down the function-means tree that was created based on the objectives and constraints discussed in item 3. Following item 4, item 5 introduces the concept generation and selection parts of the design process. There is a full morphological chart as well as the reduced morphological chart within this section. The top 5 concepts the team chose are included here as well, along with the justifications for their selections. Item 6 is a critical point within the report, as it discusses the selection of the team's top design concept. This section includes the CAD drawings of each singular part of the top concept design and follows through with an explanation of each part's purpose. Finally, Section 7 is the bill of materials that covers the materials the team plans to use on this device, along with the quantity of parts that will be used. Item 8 wraps up the entire report by summarizing the design process and concluding the findings within this report.

Prior to starting the actual design and creation of the device, the team did some background research to get a better understanding of the components they would be working with and to brainstorm potential ideas for how their device could look. The first step of research went towards making sure the team knew how to use the hardware being provided for this project. Documents from platforms such as *Arduino* provided explanations for things like circuits and motors-1] explains the wiring and coding of servo motors, along with applications to use the motors in. in finding devices that performed color sorting tasks, similar to the one the team needed their device to execute. There were many tutorials the team found in relation to color sorting devices. [2] included step-by-step instructions for how to assemble and code the devices, and these sources helped the team gain insight into the different mechanisms and components that they did not have experience using, helping brainstorm more design ideas. Another part of the team's research included looking into patents for already existing devices and considering ethical concerns that may arise through the process of designing this device. This part of the research allowed the team to find good starting points for brainstorming ideas that could potentially enable their device's functionality.

3. Problem Definition:

3a. Section Introduction :

In order to first start designing a device, there must be details on the different *design objectives*, *constraints* and *metrics* for the specific problem a device is to solve. A *design objective* can be defined as a feature or behavior that the device should have or exhibit [4]. A *constraint* is a limit or restriction on the features or behaviors of the design with the caveat that the proposed design is unacceptable if these limits are violated. *Metrics* are ways that the success of a design is measured [4]. To determine these *design objectives*, *constraints* and *metrics*, several steps had to be taken, consisting of a *Problem Statement*, *Objectives List*, *Objective Tree* and *Constraints with Metrics*. The *Problem Statement* is a short summary that contains 30 words to explain the main goals that are most important to the device. The *Problem Statement* is as follows:

To create a portable device intended to consistently sort, count, and eject ping pong balls accurately based on their color, while being market appealing.

Together, these items allow the team to understand what problem the device is intended to solve by identifying *design objectives* and *metrics* while limiting the design space [1] with *constraints*.

Table 1. List of Objectives with Metrics

Objectives & Metrics:	Explanation/Justification:
1. Durable 2. Effective/Capable 3. Sturdy/stable 4. Accurate 5. Useful 6. Aesthetically pleasing 7. Consistent 8. Fast processing 9. Eco-friendly 10. Logical design 11. Reusable 12. Operator friendly 13. Collaborative 14. Creative 15. Marketable/mass producible 16. Simple 17. Manageable	Sturdy: This objective is highly important as if the design of the device isn't sturdy, it can be broken during the stages of operation, or damaged when transported. Also, if the device isn't sturdy during operation, it can improperly sort the balls while running. The metrics to understand if the objective was met are: - <i>Number of times the device needs to be adjusted/recalibrated when in use</i> - <i>Number of parts that come loose during operation</i> Accurate: This objective is important because if the device isn't accurate then it doesn't do its major objective of sorting balls by color. Also by having an accurate device, it becomes more marketable and appealing to the audience. The metrics to understand if the objective was met are: - <i>Percent of success >95%</i> - <i>Accurate Photoreceptors readings</i> - <i>Counter Increases</i>

Table 1. List of Objectives with Metrics (continued)

Objectives & Metrics:	Explanation/Justification:
18. Repairable 19. Cooperative 20. Safe 21. Inaudible 22. Commercial 23. Debuggable 24. Comprehensive Code 25. Audience friendly 26. Explainable 27. Intuitive 28. Readable display 29. Unambiguous (the role and place of each and every component and part must be unmistakable) 30. Transparent (readily understood by the manufacturer or fabricator) 31. Autonomous	Fast processing: Having a device that is Fast Processing allows for more balls being sorted in a tight time span. Fast processing also allows for a fast set up time, making the device more marketable. Metric: - <i>Amount of balls processed in 3 minutes (<10 seconds per ball)</i> Operator Friendly: Having a device that is Operator friendly is important as if the device is hard to operate it will lead to more dissatisfaction of the device, and will make it less marketable. The metrics to understand if the objective was met are: - <i>Assembly Time under 1 minute</i> - <i>Operator and Audience Rate of Satisfaction</i> Marketable/mass producible: This objective is extremely important as it encapsulates the device as a whole and ensures that when the device is made, it meets the market requirements. The metrics to understand if the objective was met are: - <i>Inexpensive to Produce</i> - <i>Within Size Limits</i> - <i>Weight under 3lbs</i> Unambiguous: This objective may seem unnecessary but the purpose of it is to ensure that each component in the device has a purpose, and to ensure that the device as well as its process, is as streamlined as possible. The metrics to understand if the objective was met are: - <i>Lines of code needed</i> - <i>Number of electronic components</i> - <i>Number of moving parts in device</i>

3b. Table explanation:

The table above (*Table 1*) is an Objectives list, which defines what goals the device should have once it is completed. Along with the objectives of the device, they are paired with metrics to measure whether the objectives were reached at the end of the project. For this project, the major objectives that the device needs to achieve is being *Sturdy*, *Accurate*, *Fast processing*, *Operator Friendly*, *Marketable* and *Unambiguous*. These objectives are then paired with their metrics in the *Justification* portion of the figure, which explains the reasoning behind the choice of objectives.

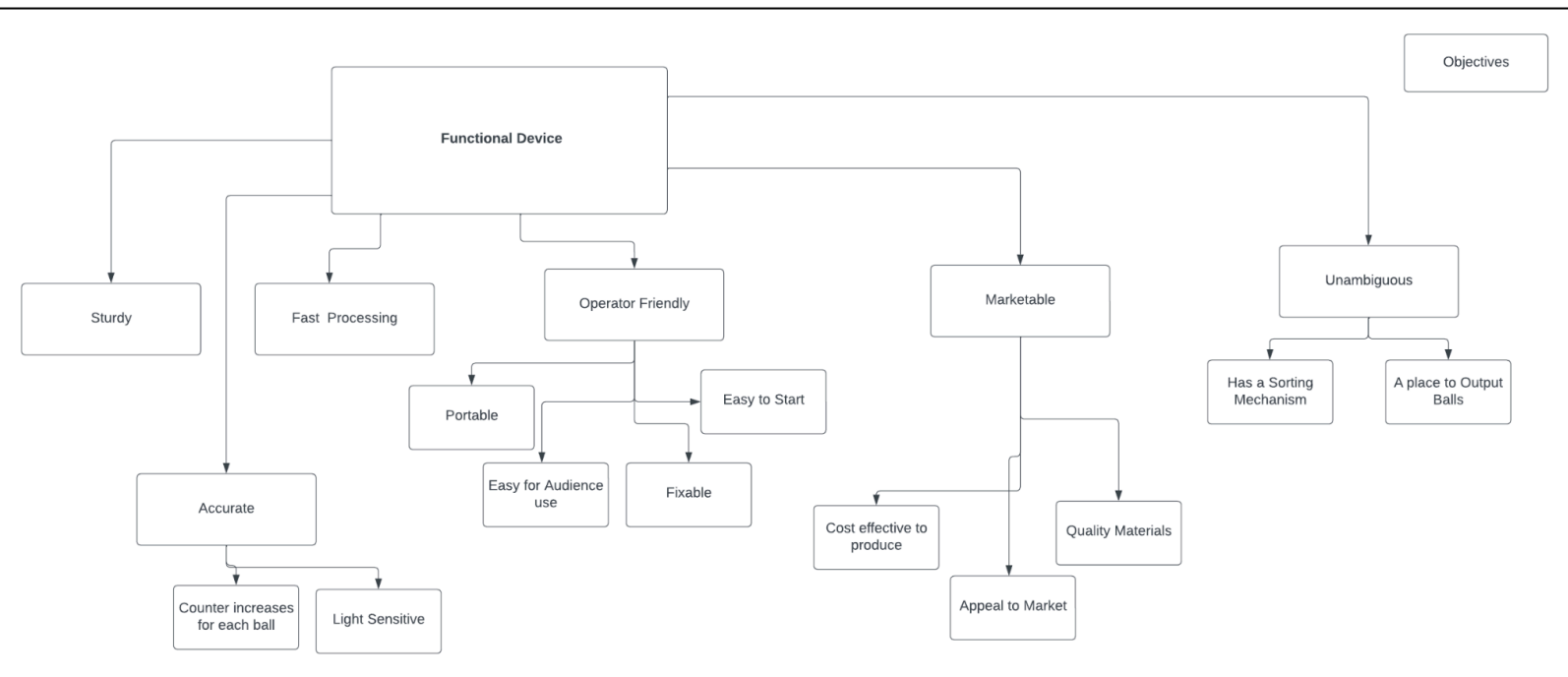


Figure 1. Objective Tree

3c. Objective tree analysis:

An Objective Tree (*Figure 1*) describes the different tasks the device will need to complete. By identifying the outputs, outcomes, and impact of the project, this tree can project the goals that the device will need to achieve during its operation. The team will keep these objectives in mind as the design process continues to ensure that the device targets all the goals set for it.

Table 2: Constraints List

Constraints	Justification For Most Important (bolded) Constraints:
1. Cost-effective/Expensive 2. Mechanical 3. Technical (code)	• Mechanical: the device should have a significant amount of moving parts. The device is not entirely dependent on code.

4. Electronic 5. Size/Portable 6. Assembly time 7. Reliable 8. Light-sensitive 9. Time efficient 10. Mobile 11. Convenient 12. No unnecessary movement 13. Not fragile 14. No part breakage during operation 15. Lightweight 16. Correctly Identifies Color 17. Counter Increases 18. Consistently recognizes balls	• Technical: the device has a significant amount of code involved, it is not only a mechanical device. • Electronic: The device has parts from one Arduino set or more, meaning there is wiring and electrical processes. • Mobile: In order to be able to transport the device when needed, it should not weigh too much. • Convenient: how satisfied the people who test the device are with the results. • Not fragile: the device should be able to run on its own without much interference during operation. • Consistently recognizes balls: the device should be able to process each ball correctly, so this is measured by the percent of correctly processed balls.
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4. Functional Analysis:

This section is about the functional analysis of the device that the team will design. Functional analysis is essentially the process of identifying what a design must do in order to be deemed as “successful”. The purpose of a functional analysis is, as stated in *Engineering Design: A Project-Based Introduction*, “to establish the functions a design must perform” [4]. This step of functional analysis is vital in the engineering design process, as it aids in making the conceptualization of designs much easier. Functional analysis helps designers develop the conceptual design of products by considering the *functions* and *means*, which allows the design team to consider all possible design choices. From this, the design choices that should and should not take place were analyzed. To see how the team organized these components, see *Figure 2*.

4a. Function-Means Tree

Dym *et al.* [4] discuss that *functions* are the things a device is designed to do, while *means* refer to a way or method in which these functions can be done. The means can be thought of as the applications through which a function can be achieved. A function-means tree acts as a “graphical representation of a design's basic and secondary functions” [4]. In this upcoming section, the team determines what *functions* are most important, and how to make the *functions* happen. This section also contains information on alternative *means* for these functions. The tree, originally shown below in *Figure 2*, has been split into 8 separate figures for viewing convenience. Each separated branch is grouped based on the top/uppermost three secondary functions that the team developed: *trigger counting*, *determine light level*, and *sort based on color*. A function-means tree is read in the following manner: the top of the tree starts at the *primary function*, listed at the top within a rectangular box. From the primary function, the tree

branches its way down into means that relate to that function. All functions under the primary function are known as *secondary functions*. This pattern of function, mean, function, etc, repeats down each level of the tree. Rectangular boxes represent functions, and trapezoids represent means.

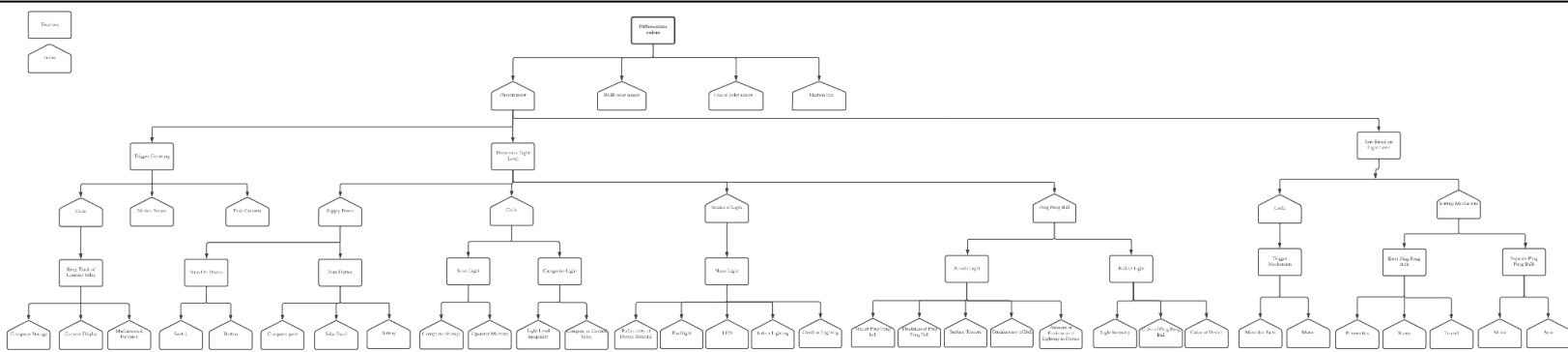


Figure 2. Whole Tree View

*Please note that not all functions or means within the tree are going to be used- this tree simply gives every possible design idea the team has evaluated.

4b. Tree Analysis:

The first level of the tree is shown below in *Figure 3*. It includes the primary function, *differentiate colors*. The primary function is the main purpose/objective the team has for their device. From the primary function, the tree moves into the second level, which are the means that correlate with the function. Notice the change in box shape. The primary function splits into 4 different means: *photoresistor*, *RGB color sensor*, *hue of color sensor*, and *human eye*. All four of these means are ways in which this primary function can be carried out. Based on set restrictions within the Project 2 assignment, the team has only branched out from the *photoresistor mean*, which is focused on in *Figure 4*.

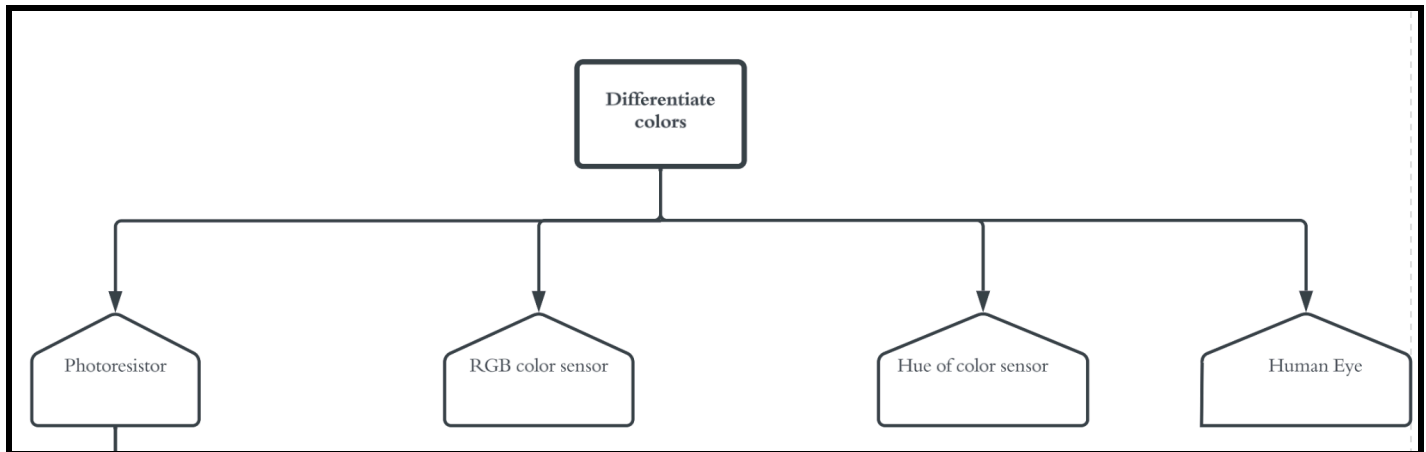


Figure 3. First Level Stating Primary Function and Means

4c. Continued Tree Analysis:

Moving down into the 2nd and 3rd levels of the tree, *Figure 4* shows the *photoresistor* mean with its secondary functions branching from it. The three functions for the photoresistor being used within the project are *trigger counting*, *determine light level*, and *sort based on light level*. These secondary functions are required for the chosen mean.

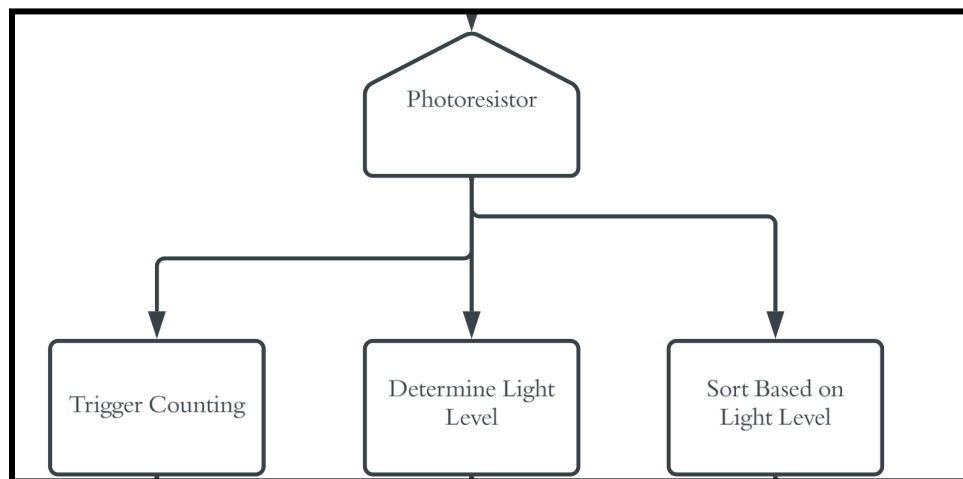


Figure 4. Selected *Photoresistor* Mean with Objectives

4d. Continued Tree Analysis:

Continuing is the branch relating to *trigger counting*, one of the 3 functions of the photoresistor within the device. The means under trigger counting include ways in which counting could potentially be triggered in the device. All three of the listed means can trigger

counting, but only one *code* will actually be used as a component within the device, as per design rules. The *code* then flows down into its' secondary function: the device being able to keep track of the counter value. There are 3 means associated with this secondary function, described below in *Figure 5* as *computer storage*, *counter display* and *mathematical function*.

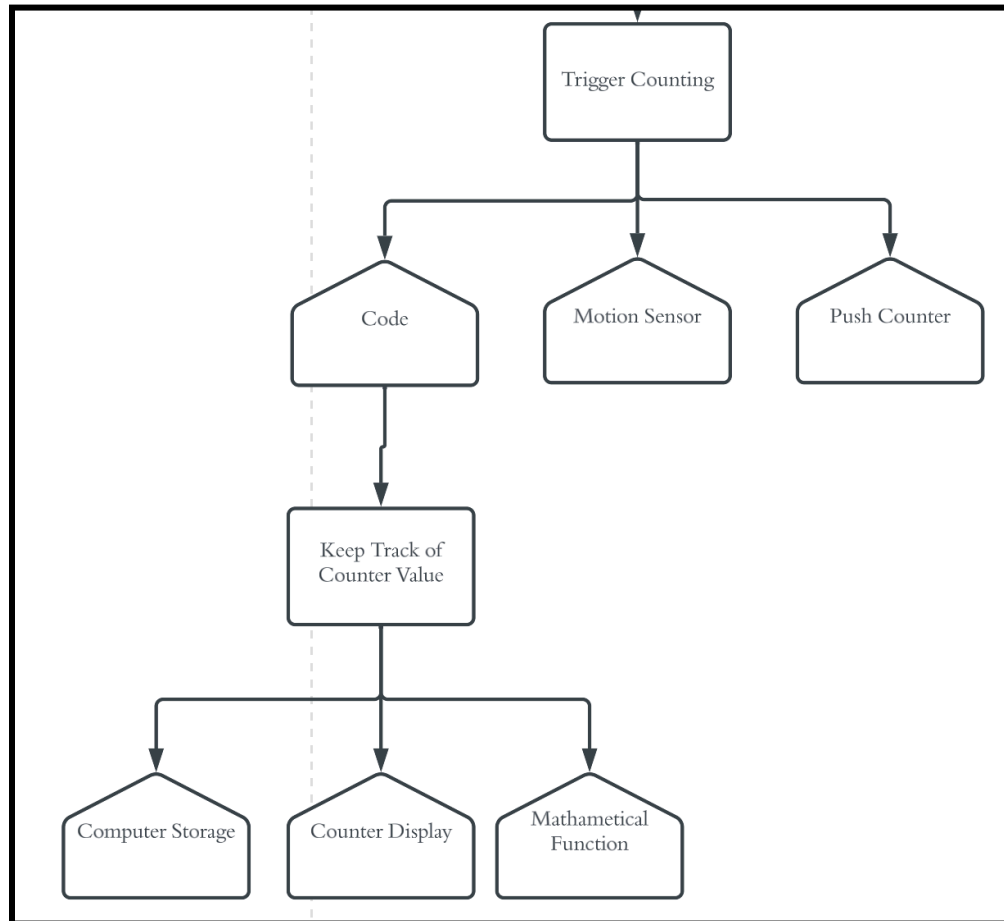


Figure 5. Functions and Means under *Trigger Counting* Branch

4e. Continued Tree Analysis:

The function of *determining light level* can be found below within *Figures 6-9*, split up into four images for reading convenience. In *Figure 6* (1 of 4), the mean explored under *determining light level* is *supply power*. A power supply is necessary in having the photoresistor read the light level it is receiving and provide an accurate decision based on the readings. From there, the means go back into more secondary functions. In order to determine light level, power must be supplied. With the power being supplied, the device needs to be able to be turned on and it needs to be able to run. To turn on the device, there should be either a switch or a button.

Similarly, running the device will take place through the use of a computer port, solar panel, and/or battery.

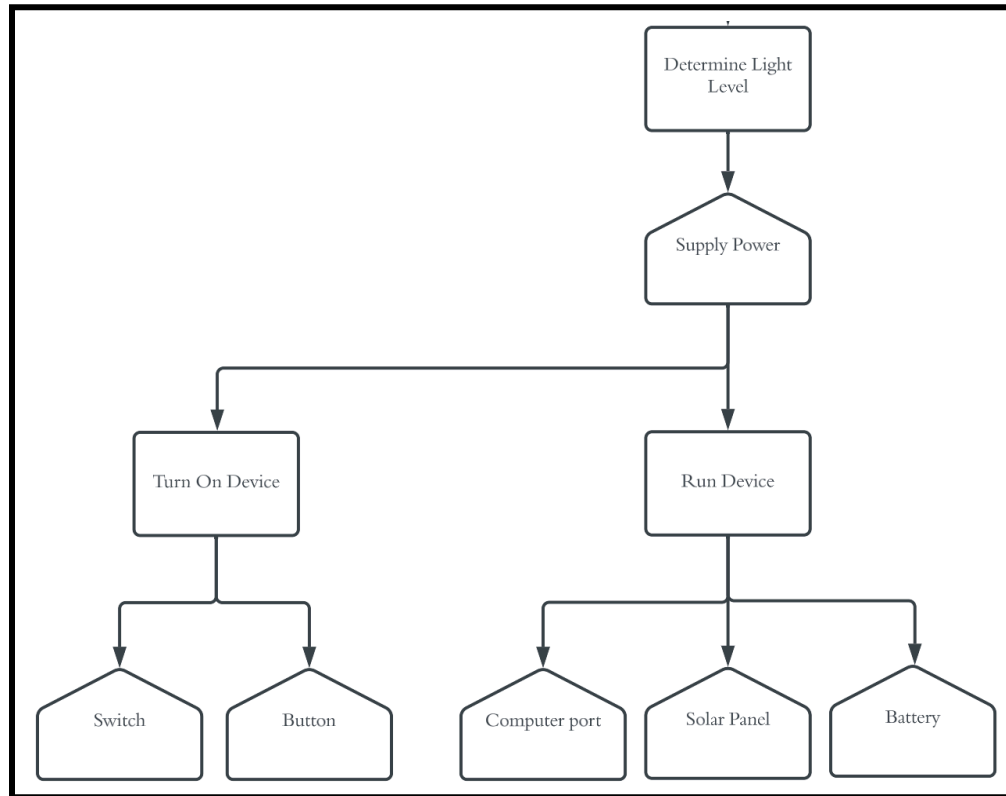


Figure 6. Means and Functions under *Determine Light Level* Branch (1 of 4)

4f. Continued Tree Analysis:

Figure 7 (2 of 4) demonstrates a similar concept, but with the mean for determining the light level now moving to having *code*. Through the code, the functions of storing and categorizing light will be accomplished. From the code being able to store light in order to determine the light level, computer storage and operator memory are both means under this. Means for categorizing light include *light level inequalities* (written into the code), and *compare to control value* (also within the code).

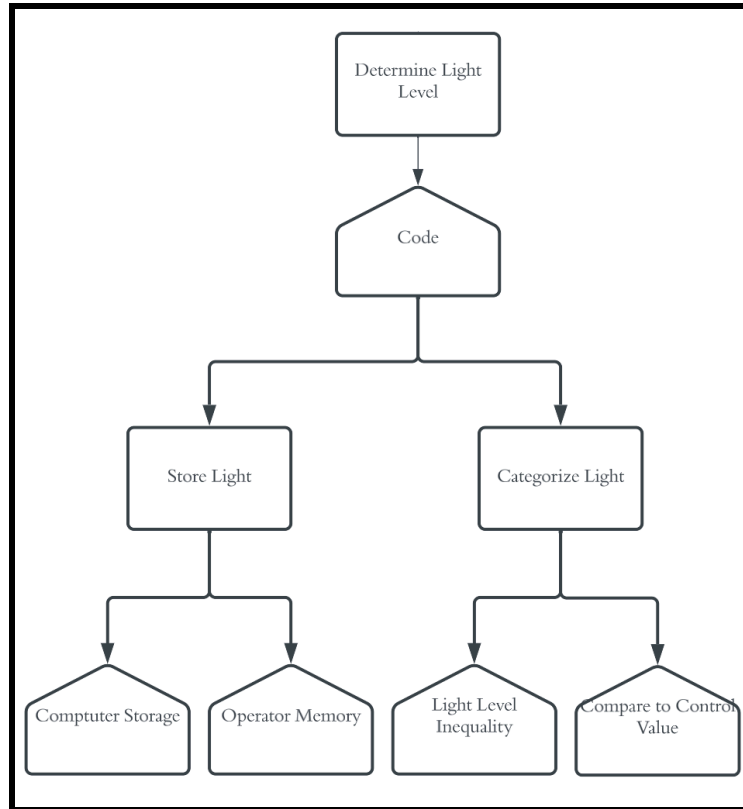


Figure 7. Means and Functions under *Determine Light Level* Branch (2 of 4)

4g. Continued Tree Analysis:

The next means for determining light level is having a source of light. This mean's branch, *Figure 8* (3 of 4), defines a secondary function for the source of light to *shine light*. From here, 5 separate means are developed: *reflectivity of the device material, flashlight, LED, indoor lighting, and outdoor lighting*.

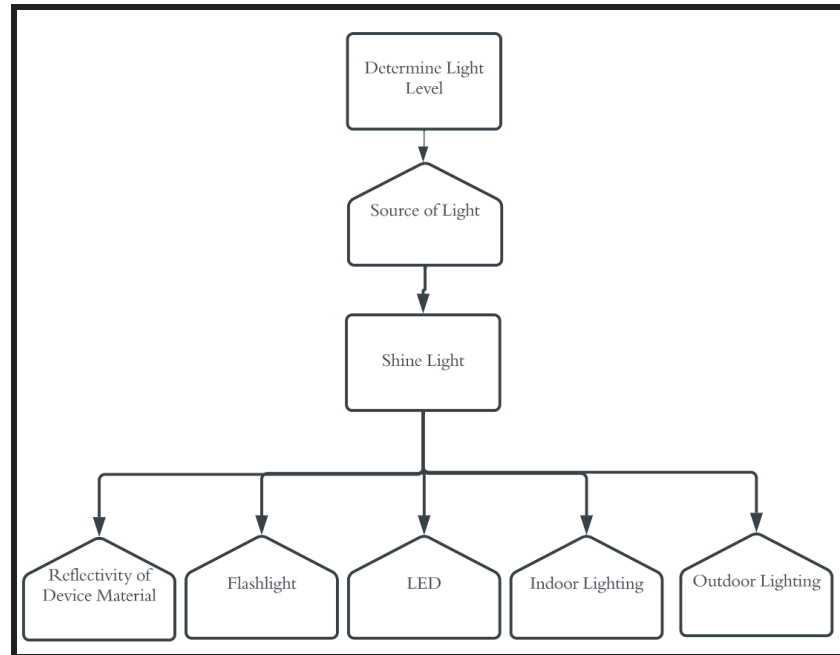


Figure 8. Means and Objectives under *Determine Light Level* Branch (3 of 4)

4h. Continued Tree Analysis:

The last branch (4 of 4) under *Determine Light Level* relates to the ping pong balls themselves. To differentiate the colors of the balls, the ping pong balls themselves must be present. These ping pong balls will both absorb and reflect light, which would be the secondary functions under their mean. Absorbing the light would include size, thickness, texture, and translucency of the balls. In terms of *reflecting light, the light intensity, color of the ping pong ball, and color of the device* are needed.

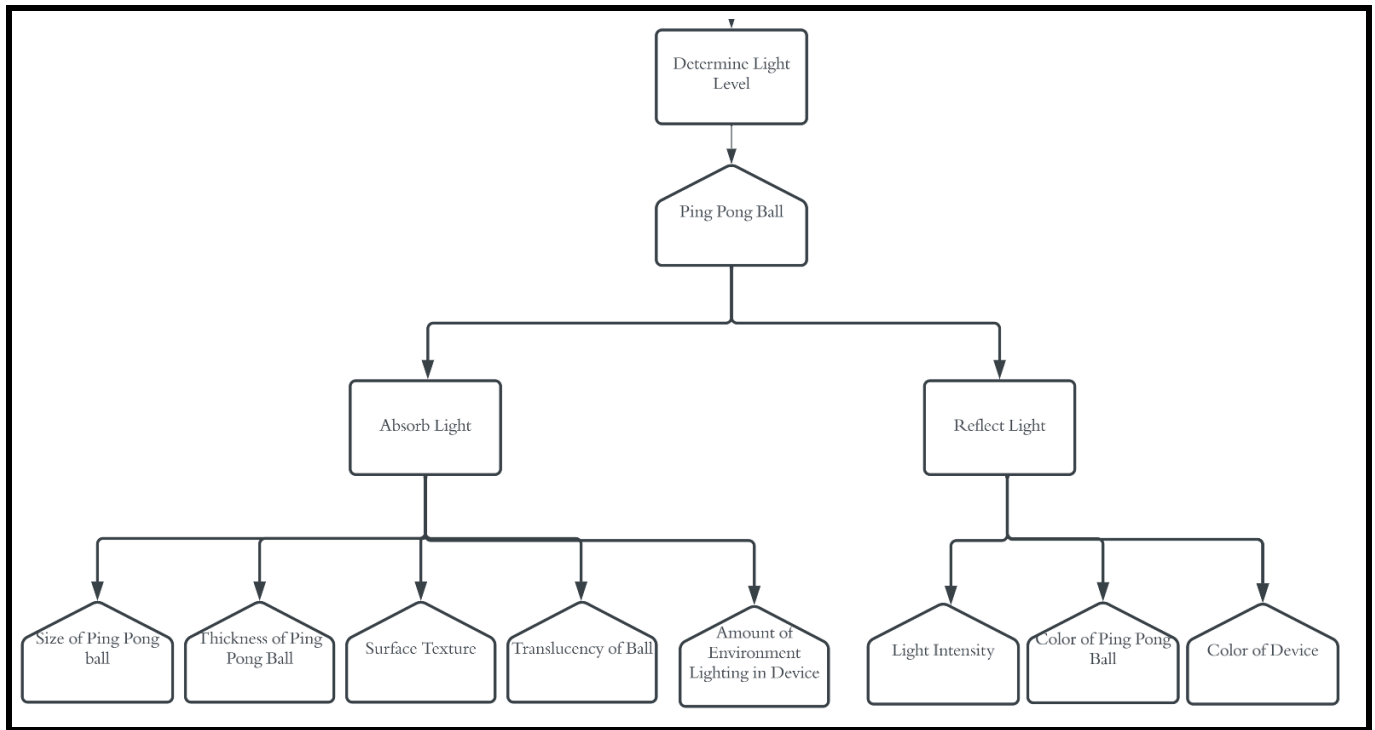


Figure 9. Means and Objectives under *Determine Light Level* Branch (4 of 4)

4i. Continued Tree Analysis:

The team lastly discussed how the photoresistor would work to *Sort Based on Light Level*. The means for sorting based on the light level are *code* and a *sorting mechanism*. The code is going to be what sets off the mechanism and gets it to move, hence the function *trigger mechanism*. In order to have the code trigger the mechanism, it will require both *movable parts* and a *motor*. In terms of the *sorting mechanism*, it will need to eject the ping pong balls as well as separate them. In order to eject the ping pong balls, the following means will be needed: pneumatics, a ramp, and/or a funnel. These three are means to accomplish the task of ejecting the ping pong balls. Separating them will require a motor and some sort of mechanical arm that will be the actual separator. This process can be seen in *Figure 10* below.

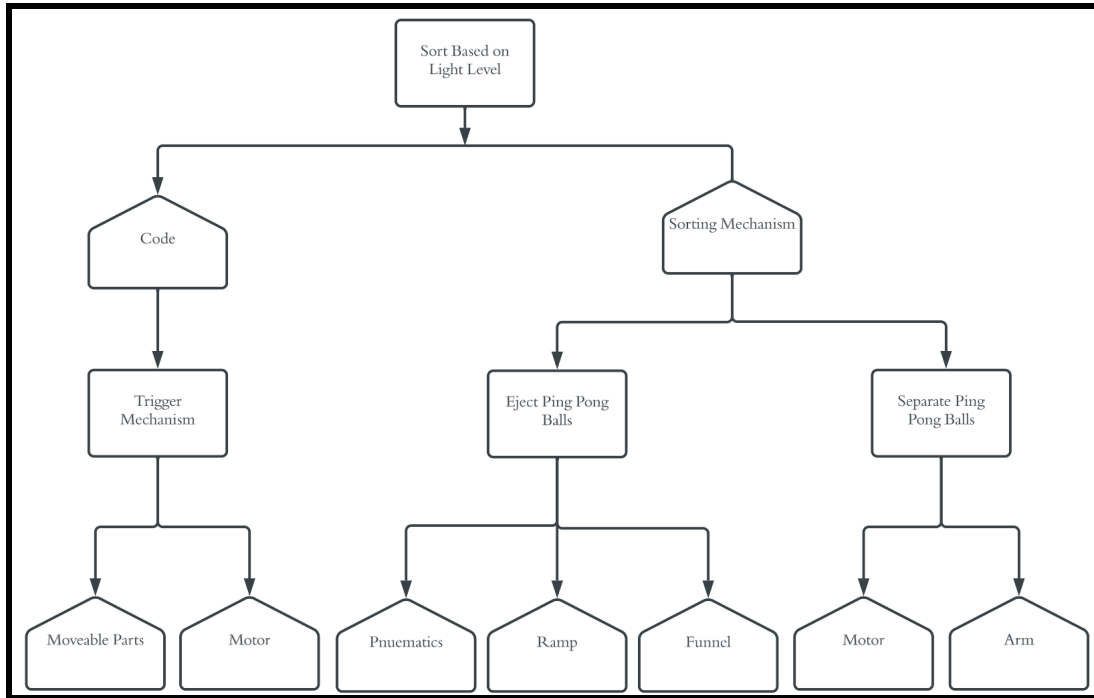


Figure 10. Remaining Objectives and Means under *Sort Based on Light Level* Branch

5. Concept Generation and Selection Introduction:

5a. Section Introduction:

This section is about how the team generated different concepts for the design space of the device that will be built. A design space is a space where important parameters and desired attributes of the device are brought together to help the designers finalize the design of the device. It can be used to help designers generate potential design alternatives for a design problem. A concept is a design solution, or a combination of the different means from each function, so looking at the different possible concepts allows the team to consider all possible ways to create the desired device. *Engineering Design: A Project-Based Introduction* explains how the morph chart plays a major role in helping a design team select a final design. As explained in the text, functions within a morph chart are always to be noted as noun-verb action pairs [4].

There are a few items in this section that require explanation. Below this paragraph there is a morphological chart (Tables 3 & 4). This chart allows the team to visualize the design space for this by placing the functions and their associated means in a table. This is what creates the design space. A design space is “an imaginary intellectual region of design alternatives that contains all of the potential solutions to a design problem” [4]. The morph chart ultimately outputs this total number of possible design choices based on the number of means. For example, having an additional mean would double the total number of possible design

choices, because all of the means are multiplied by one another to form the design space [4]. After making the morphological chart (table 5), the team then worked to reduce it (see item 5c for explanation of the process). With these reduced means, the team was able to combine them to make 20 concepts as listed in tables 6, 7, and 8 of item 5e. This list was created by making as many possible unique combinations that had no repeating means. Note that this list does not include every possible concept, only 20 that the team deemed promising. To see a justification and further explanation of this list, see item 5d. Next in this section is a list of the top 5 concepts, (table 9) which has been derived from the previous list of 20 concepts. The reasoning behind how the team chose these five concepts are explained in the next item, justified list of top concepts (see item 5f, table 10).

5b. Morphological chart(s) explanations:

A morphological chart, often referred to as a morph chart is used to chart out and recognize the different means that can be put towards getting certain functions to happen. The team has given the whole morph chart in Table A. The chart starts with all of the functions formulated in the function-means tree ordered and numbered in the first two columns. Every column after the named functions deals with separate means that go with the corresponding function. These means were brainstormed and produced in the function-means tree, and are simply transferred into the morph chart at this point in the concept generation process. Note the value highlighted at the bottom right corner: this number gives the total possible number of design solutions that can be created based on the means given in the morphological chart. A large design space indicates a large number of design problems with a large number of possible design variables. This design space (the highlighted number) is calculated through the multiplying the total number of means by one another (multiply all total # of mean values in part 2 of table 5 by one another.)

Table 3. Full Morphological Chart Part 1 (first half of the table with means 1&2)

Functions	Functions Names	Means 1	Means 2	Total # of Means
Function 1	<i>Differentiate Colors</i>	Photoresistor	RGB Color Sensor	1
Function 2	<i>Trigger Counting</i>	code	motion sensor	1
Function 3	<i>Determine Lightlevel</i>	Ping Pong Ball	Supply Power	1
Function 4	<i>Sort based on Light Level</i>	code	sorting mechanism	1
Function 5	<i>Keep Track of Counter Value</i>	Computer Storage	Counter Display	1
Function 6	<i>Turn on device</i>	Switch	Button	1
Function 7	<i>Power device</i>	Computer Port	Solar Power	1
Function 8	<i>Categorize the Light Levels</i>	Light level Inequalities	compare to control value	1

Function 9	<i>Shine Light</i>	Indoor Lighting	Flashlight	1
Function 10	<i>Store Light Level</i>	Computer Storage	Operator Memory	1
Function 11	<i>absorb light</i>	translucency of ball	surface texture	1
Function 12	<i>reflect light</i>	light intensity	color of ping pong ball	1
Function 13	<i>trigger mechanics</i>	movable parts	motor	1
Function 14	<i>eject ping pong balls</i>	ramp	funnel	1
Function 15	<i>separate ping pong balls</i>	motor	arm	1
				6,220,800

Table 4. Full Morphological Chart Part 2 (second half of table with means 3,4, & 5)

Functions	Function Names	Means 3	Means 4	Means 5	Total # of Means
Function 1	<i>Differentiate Colors</i>	Hue of Color Sensor	Human Eye		4
Function 2	<i>Trigger Counting</i>	push counter			3
Function 3	<i>Determine Lightlevel</i>	Code			4
Function 4	<i>Sort based on Light Level</i>				2
Function 5	<i>Keep Track of Counter Value</i>	mathematical function			3
Function 6	<i>Turn on device</i>				2
Function 7	<i>Power device</i>	Battery			3
Function 8	<i>Categorize the Light Levels</i>				2
Function 9	<i>Shine Light</i>	LED	reflective material of device	Outdoor Lighting	5

Function 10	<i>Store Light Level</i>				2
Function 11	<i>absorb light</i>	thickness of ping pong ball	size of ping pong ball	amount of environment lighting let into device	5
Function 12	<i>reflect light</i>	color of device			3
Function 13	<i>trigger mechanics</i>				2
Function 14	<i>eject ping pong balls</i>	pneumatics			3
Function 15	<i>separate ping pong balls</i>				2
					6,220,800

5c. Reduced Morphological Chart explanation:

The team's reduced morph chart is given below as Table 5. A reduced morphological chart is essentially taking the whole version and getting rid of the means that are not possible/within reach for the project's design. In order to reduce the full morph chart, the team went through each function, checking if every mean met the project requirements. Following this was the removal of the means that the team did not see as being part of their actual design. Many of the means the team had to remove from the full morph chart for Table 5, were due to restrictions given within the project description and rubric. For example, in Project 2, a photoresistor must be what helps determine the color of ping pong balls, so having an RGB color sensor or a hue of color sensor in the device would not be able to satisfy the requirements. Lastly, it is important to note the highlighted number at the bottom right corner of Table 5. This is the new (reduced) number of possible design solutions based on the means now available.

Table 5 . Reduced Morphological Chart Part 1 (first part of table)

Functions	Function Names	Means 1	Means 2	Total # of Means
Function 1	<i>Differentiate Colors</i>	Photoresistor		1
Function 2	<i>Trigger Counting</i>	code		1
Function 3	<i>Determine Lightlevel</i>	Ping Pong Ball	Supply Power	1
Function 4	<i>Sort based on Light Level</i>	code	sorting mechanism	1

Function 5	<i>Keep Track of Counter Value</i>	Computer Storage	Counter Display	1
Function 6	<i>Power Device</i>	Computer Port	Battery	1
Function 7	<i>Turn on device</i>	Switch	Button	1
Function 8	<i>Categorize the Light Levels</i>	Light level Inequalities		1
Function 9	<i>Shine Light</i>	Indoor Lighting	LED	1
Function 10	<i>Store Light Level</i>	Computer Storage		1
Function 11	<i>absorb light</i>	translucency of ball	surface texture	1
Function 12	<i>reflect light</i>	light intensity	color of ping pong ball	1
Function 13	<i>trigger mechanics</i>	motor		1
Function 14	<i>eject ping pong balls</i>	ramp	funnel	1
Function 15	<i>separate ping pong balls</i>	motor	arm	1
				23,040

Table 5 . *Reduced Morphological Chart Part 2 (continued)*

Functions	Function Names	Means 3	Means 4	Total # of Means
Function 1	<i>Differentiate Colors</i>			1
Function 2	<i>Trigger Counting</i>			1
Function 3	<i>Determine Lightlevel</i>			1
Function 4	<i>Sort based on Light Level</i>	Code	Source of Light	1
Function 5	<i>Keep Track of Counter Value</i>			1
Function 6	<i>Power Device</i>	mathematical function		1
Function 7	<i>Turn on device</i>			1
Function 8	<i>Categorize the Light Levels</i>			1

Function 9	<i>Shine Light</i>			1
Function 10	<i>Store Light Level</i>	reflective material of device	Outdoor Lighting	1
Function 11	<i>absorb light</i>			1
Function 12	<i>reflect light</i>	thickness of ping pong ball	size of ping pong ball	1
Function 13	<i>trigger mechanics</i>			1
Functions	Function Names	Means 1	Means 2	Total # of Means
Function 14	<i>eject ping pong balls</i>			1
Function 15	<i>separate ping pong balls</i>	pneumatics		1
				23,040

5d. Justification of Top 20 concepts & Design Decisions:

The team's top 20 concepts and designs were based on the *means* and *functions* from the *morph chart* and reduced *morph chart*. The concepts were each unique to their own, and it resulted in the choice of which would be the most relevant to our final design. All of the concepts consisted of a *photoresistor*, *code*, *motor*, and *arm*. The rest of the *functions* had differing *means*. The team came up with different combinations, but stuck with those four *means* in each format. By keeping that in mind, the team also knew that the *means* that were discussed in the rubric had to be included. For example, a *photoresistor* had to be included, as it is a requirement based on the rubric. For the remainder of the *functions*, the team proceeded to insert the relevant *means* for each of them. In *functions* 2, 10, 13, and 15, these were the best means based on the reduced *morphological chart*. Then, the team decided to add those *means* into each concept based on the ideas of each member.

5e. List of Top 20 Concepts:

Table 6: Top 20 Concepts (functions 1 through 5)

	<i>Function 1</i>	<i>Function 2</i>	<i>Function 3</i>	<i>Function 4</i>	<i>Function 5</i>
Concept 1	Photoresistor	Code	Ping Pong Ball	sorting mechanism	Counter Display
Concept 2	Photoresistor	Code	Supply Power	sorting mechanism	Counter Display

Concept 3	Photoresistor	Code	Source of Light	sorting mechanism	Counter Display
Concept 4	Photoresistor	Code	Ping Pong Ball	sorting mechanism	Counter Display
Concept 5	Photoresistor	Code	Supply Power	sorting mechanism	Counter Display
Concept 6	Photoresistor	code	Source of Light	sorting mechanism	Counter Display
Concept 7	Photoresistor	code	Ping Pong Ball	sorting mechanism	Counter Display
Concept 8	Photoresistor	code	Supply Power	sorting mechanism	Counter Display
Concept 9	Photoresistor	code	Source of Light	sorting mechanism	Counter Display
Concept 10	Photoresistor	code	Ping Pong Ball	sorting mechanism	Counter Display
Concept 11	Photoresistor	code	Supply Power	sorting mechanism	mathematical function
Concept 12	Photoresistor	code	Source of Light	sorting mechanism	mathematical function
Concept 13	Photoresistor	code	Ping Pong Ball	sorting mechanism	mathematical function
Concept 14	Photoresistor	code	Supply Power	sorting mechanism	mathematical function
Concept 15	Photoresistor	code	Source of Light	sorting mechanism	mathematical function
Concept 16	Photoresistor	code	Ping Pong Ball	sorting mechanism	mathematical function
Concept 17	Photoresistor	code	Supply Power	sorting mechanism	mathematical function
Concept 18	Photoresistor	code	Source of Light	sorting mechanism	mathematical function
Concept 19	Photoresistor	code	Ping Pong Ball	sorting mechanism	mathematical function

Concept 20	Photoresistor	code	Supply Power	sorting mechanism	mathematical function
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Table 7: list of top 20 concepts (functions 6-10)

	Function 6	Function 7	Function 8	Function 9	Function 10
Concept 1	Computer Port	Switch	Light level Inequalities	indoor light	Computer Storage
Concept 2	Battery	Switch	Light level Inequalities	LED	Computer Storage
Concept 3	Computer Port	Switch	Light level Inequalities	reflective material of the device	Computer Storage
Concept 4	Battery	Switch	Light level Inequalities	outdoor lighting	Computer Storage
Concept 5	Computer Port	Switch	Light level Inequalities	indoor light	Computer Storage
Concept 6	Battery	Switch	Light level Inequalities	LED	Computer Storage
Concept 7	Computer Port	Switch	Light level Inequalities	reflective material of the device	Computer Storage
Concept 8	Battery	Switch	Light level Inequalities	outdoor lighting	Computer Storage
Concept 9	Computer Port	Switch	Light level Inequalities	indoor light	Computer Storage
Concept 10	Battery	Switch	Light level Inequalities	LED	Computer Storage
Concept 11	Computer Port	Button	Light level Inequalities	reflective material of the device	Computer Storage
Concept 12	Battery	Button	Light level Inequalities	indoor light	Computer Storage
Concept 13	Computer Port	Button	Light level Inequalities	outdoor lighting	Computer Storage

Concept 14	Battery	Button	Light level Inequalities	LED	Computer Storage
Concept 15	Computer Port	Button	Light level Inequalities	reflective material of the device	Computer Storage
Concept 16	Battery	Button	Light level Inequalities	LED	Computer Storage
Concept 17	Computer Port	Button	Light level Inequalities	indoor light	Computer Storage
Concept 18	Battery	Button	Light level Inequalities	LED	Computer Storage
Concept 19	Computer Port	Button	Light level Inequalities	outdoor lighting	Computer Storage
Concept 20	Battery	Button	Light level Inequalities	reflective material of the device	Computer Storage

Table 8: list of top 20 concepts(functions 11-15)

	Function 11	Function 12	Function 13	Function 14	Function 15
Concept 1	translucency of ball	Light intensity	motor	ramp	arm
Concept 2	translucency of ball	color of ping pong ball	motor	ramp	arm
Concept 3	translucency of ball	Light intensity	motor	ramp	arm
Concept 4	translucency of ball	color of ping pong ball	motor	ramp	arm
Concept 5	size of ping pong ball	Light intensity	motor	ramp	arm
Concept 6	size of ping pong ball	color of ping pong ball	motor	ramp	arm
Concept 7	size of ping pong ball	Light intensity	motor	funnel	arm

	<i>Function 11</i>	<i>Function 12</i>	<i>Function 13</i>	<i>Function 14</i>	<i>Function 15</i>
Concept 8	size of ping pong ball	color of ping pong ball	motor	funnel	arm
Concept 9	surface texture	Light intensity	motor	funnel	arm
Concept 10	surface texture	color of ping pong ball	motor	funnel	arm
Concept 11	surface texture	Light intensity	motor	funnel	arm
Concept 12	surface texture	Light intensity	motor	ramp	arm
Concept 13	darkness of ping pong ball	color of ping pong ball	motor	funnel	arm
Concept 14	darkness of ping pong ball	color of ping pong ball	motor	funnel	arm
Concept 15	darkness of ping pong ball	Light intensity	motor	pneumatics	arm
Concept 16	darkness of ping pong ball	color of ping pong ball	motor	ramp	arm
Concept 17	environment lighting let into device	Light intensity	motor	pneumatics	arm
Concept 18	environment lighting let into device	color of ping pong ball	motor	funnel	arm
Concept 19	environment lighting let into device	color of ping pong ball	motor	pneumatics	arm
Concept 20	environment lighting let into device	light intensity	motor	pneumatics	arm

5f. Justification of Top 5 Concepts:

Following our top 20 concepts, was the selection of our top 5 concepts. These include Concept 12, Concept 14, Concept 16, Concept 18, and Concept 20. The list of the top 5 concepts can be shown on *Table 10* below. To further reduce the options, the team followed what was

wanted in our design, instead of what was necessary. As a result, the team knew a *photoresistor*, including *code*, a *motor*, and an *arm* needed to be included in the final design. The team chose the top 5 concepts based on the means that they consisted of. Each concept is a different combination of means, so the team looked at the 20 top combinations and chose five that had preferable means.

Table 9. *Top Five Concepts*

Top 5 Concepts:	Means associated with each concept:
Concept 12	Photoresistor, code, Source of Light, sorting mechanism, mathematical function, batter, button, light level inequality, indoor light, computer storage, surface texture, light intensity, motor, ramp, arm,
Concept 14	Photoresistor, code, Supply Power, sorting mechanism, mathematical function, Battery, Button, Light level Inequalities, LED, Computer Storage, darkness of ping pong ball, color of ping pong ball, motor, funnel, arm,
Concept 16	Photoresistor, code, Ping Pong Ball, sorting mechanism, mathematical function, Battery, Button, Light level Inequalities, LED, Computer Storage, darkness of ping pong ball, color of ping pong ball, motor, ramp, arm
Concept 18	Photoresistor, code, Source of Light, sorting mechanism, mathematical function, Battery, Button, Light level Inequalities, LED, Computer Storage, environment lighting let into device, color of ping pong ball, motor, funnel, arm
Concept 20	Photoresistor, code, Supply Power, sorting mechanism, mathematical function, Battery, Button, Light level Inequalities, reflective material of the device, Computer Storage, environment lighting let into device, light intensity, motor, pneumatics, arm

6. Concept Selection:

Oftentimes in the design process, individuals have to compare their ideas and compile the best possible solution using these comparisons. A comparison chart makes this concept comparison and selection process a lot smoother; a comparison chart simply allows the team to visualize the different viable concepts, and which options meet certain criteria better [4]. The criteria for this specific comparison chart consist of five objectives and five constraints that the team deemed most important. Whether the concept meets these criteria or not is illustrated through the priority check mark method. The way this method of comparison works is that objectives or constraints that are a top priority are given three check marks, the least prioritized ones are given one check mark, and the ones in the middle are given two. These top five concepts are then ranked by checkmarks. For more explanation, see item 6a.

6a. Section Introduction/Explanation:

In order to narrow down the design options, the team utilized the Priority Check Mark method (*Table 11*), which qualitatively reflects a client's values in terms of the priority assigned to each objective [4]. This method was selected because of its simple and qualitative organization, which makes it easy to understand and display. The first step of this process was to input five of the most important objectives and five of the most important constraints from the Objective-Constraint Tree. These are listed on the left of the table above and are labeled with (O) for Objective or (C) for Constraint. These were chosen based upon the design requirements from the Design Rules document, and they were discussed with the team based upon importance. Under the *Priority* headline, there are a number of checkmarks. Three check marks signify the objective/constraint that is extremely important whereas 1 check mark signifies a lesser important objective/constraint. These checkmarks are then listed across the table for each concept. Also labeled in each concept, is a *1x* or *0x*. These numbers signify whether the concept meets the design objective/constraint. The concept with the most number of *1x*'s was the winning concept as it upheld the design objectives/constraints.

Concept 16 (the highlighted column) is the team's chosen "top concept". This was decided through the use of the comparison chart and priority check mark method previously described. These comparison techniques helped the team in choosing concept 16 as the top concept because they aided in narrowing down the most important priorities. The best design concept was much easier to choose when everything was organized and prioritized as shown below in Figure 1. Concept 16 had the most (all) *1x*'s, therefore fulfilling the design objectives and constraints best compared to the other four choices.

Table 10. Priority Check Mark Method [4]

Design Constraints & Objectives:	Priority:	Concept 12:	Concept 14:	Concept 16:	Concept 18:	Concept 20:
1. Durable (O)	✓✓✓	1x✓✓✓	1x✓✓✓	1x✓✓✓	1x✓✓✓	0x✓✓✓
2. Accurate (O)	✓✓✓	1x✓✓✓	1x✓✓✓	1x✓✓✓	1x✓✓✓	0x✓✓✓
3. Fast Processing (O)	✓✓	1x✓✓	1x✓✓	1x✓✓	1x✓✓	1x✓✓
4. Operator Friendly (O)	✓✓	1x✓✓	1x✓✓	1x✓✓	1x✓✓	0x✓✓
5. Marketable(O)	✓✓✓	1x✓✓✓	1x✓✓✓	1x✓✓✓	1x✓✓✓	0x✓✓✓
6. Cost-efficiency (C)	✓✓	1x✓✓	1x✓✓	1x✓✓	1x✓✓	0x✓✓
Design Constraints & Objectives:	Priority:	Concept 12:	Concept 14:	Concept 16:	Concept 18:	Concept 20:
7. Portability(C)	✓✓	0x✓✓	1x✓✓	1x✓✓	1x✓✓	0x✓✓
8. Reliable(C)	✓✓	1x✓✓	0x✓✓	1x✓✓	0x✓✓	1x✓✓
9. Light-Sensitive(C)	✓✓✓	1x✓✓✓	1x✓✓✓	1x✓✓✓	0x✓✓✓	1x✓✓✓
10. Convenient(C)	✓	0x✓	1x✓	1x✓	1x✓	0x✓

6b. Top Design Concept:

As stated above, Concept 16 checked all the objects and constraints deemed most important to this project (shown in *Table 12*). This concept also has features that were favorable to the team, such as involving the use of a battery as a powersource in the design so that it can be separate from the computer. This helps to make the device more portable and easily accessible. It uses an additional LED lightsource to ensure the light level of the photoresistor is accurate and not impacted by surrounding light. The usage of a ramp for an exit was also key as it ensures for a fast delivery to the bins. When designing the drawings of concept 16, there were many iterations, but the team landed on this design because of its sturdy base and user-friendly layout. In total, there are only 6 parts to design, keeping it simple and cost effective.

6c. Proof of Concept:

The team's top three designs are somewhat similar, but there are a few distinctions worth discussing. The following three concept sketches are rough ideas of what the team has come up with for logical and buildable devices.

Concept 12 Sketch

The first design, concept 12, is focused on horizontal movement. The sketch is shown directly below:

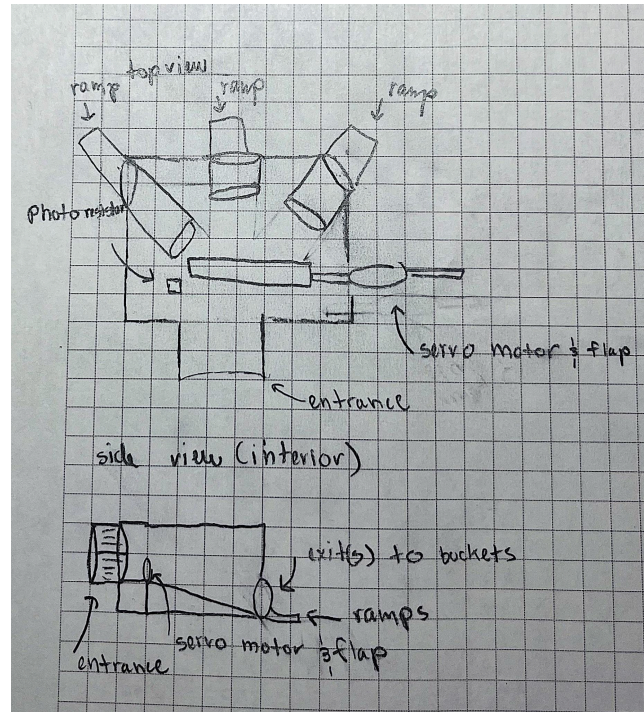


Figure 11: Rough Sketch of Concept 12

The design above would allow a ball to move horizontally on a slight slope to allow easy movement. The ball would be guided by a motor arm based on the color reading. It would be allowed into one of the three tunnels. This design did not become the final design because of concerns relating to the funnel guiding the ball into the correct tunnel and the fact that the horizontal design wouldn't work with gravity very well, so it would allow for more errors.

Concept 14 Sketch

The second of the top designs is of concept 14. The sketch is shown below:

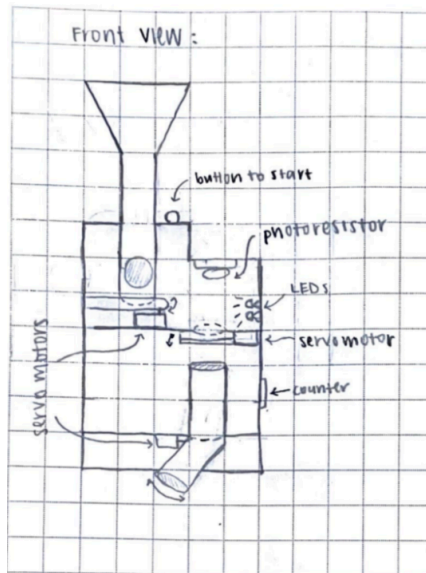


Figure 12: Rough Sketch of Concept 14

This design has a more vertical design, so it works with gravity more efficiently than the previous. The way this design would work is that after the ball enters the device, and after having its color identified, a tubular arm would move it appropriately. This design would have been more difficult to manufacture, mostly because of the tubular arm and because the ball would not exactly fall down a straight vertical path. There would be too many potential mechanical errors.

Concept 16 Sketch

The last of the top three, concept 16, was considered the best due to its simple yet effective design. This is the team's chosen design concept for their device. The image is shown below:

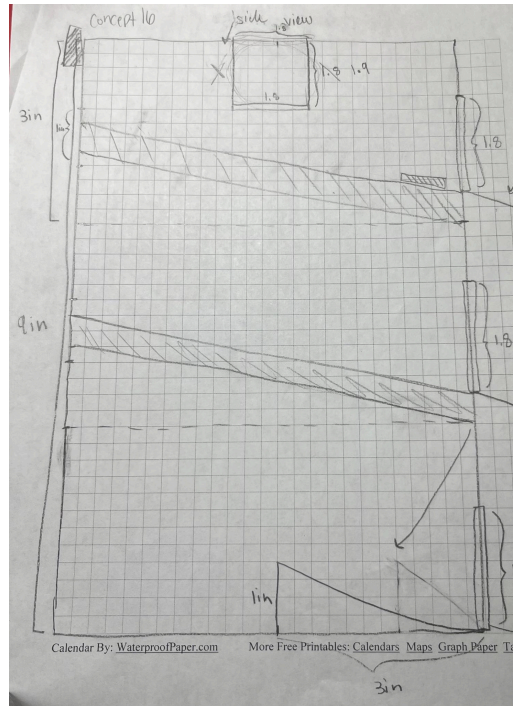


Figure 13: *Rough Sketch of Concept 16 (winning concept)*

This design consists of dropping ramps above one another in a rectangular frame, and the balls are guided through openings in the frame. There will also be outer ramps to guide the balls into the correct buckets.

6c. Final Design Section:

This next section contains all of the engineering CAD drawings for each separate part of the final design, as well as the full CAD model assembly*. The drawings are turned horizontally to make dimensions easier to read. The order of the drawings is listed below and are ordered in the way the device operates:

1. All pieces are attached to the *Frame* (Figure 14 on page 32) of the device which is square in shape and will provide stability to the entire device.

2. Ball is inserted into *Funnel* piece and the gateway (shown as the cut in the side of the tube) will stop the ball from entering the main device, where the color will be determined from the photoresistor (*Figure 15 on page 33*).
3. The *Interior Ramps* (*Figure 16 on page 34*) will move according to which exit the ball needs to be sorted.
4. The *Bottom Interior Ramp* (*Figure 17 on page 35*) will act as a segway for the ball's last exit.
5. Depending on which exit the ball is sorted to, *outer ramp 1* (*Figure 18 on page 36*) or *outer ramp 2* (*Figure 19 on page 35*) will guide the ball to the correct bin.

**Figure 20* on page 36 shows the assembled CAD model of the final device design that is based off of the winning concept.

Figure 14: CAD Frame

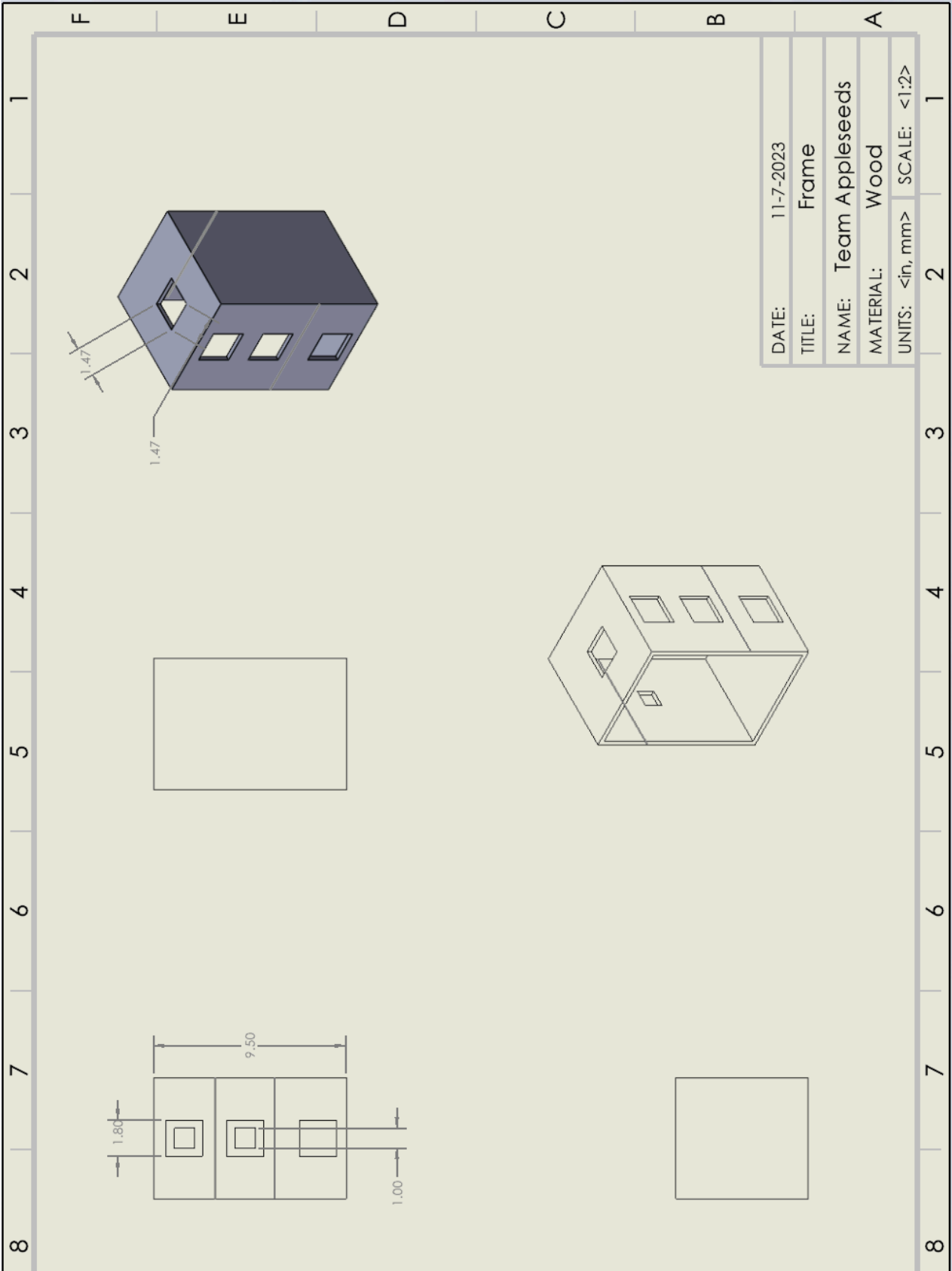


Figure 15: CAD Funnel Piece

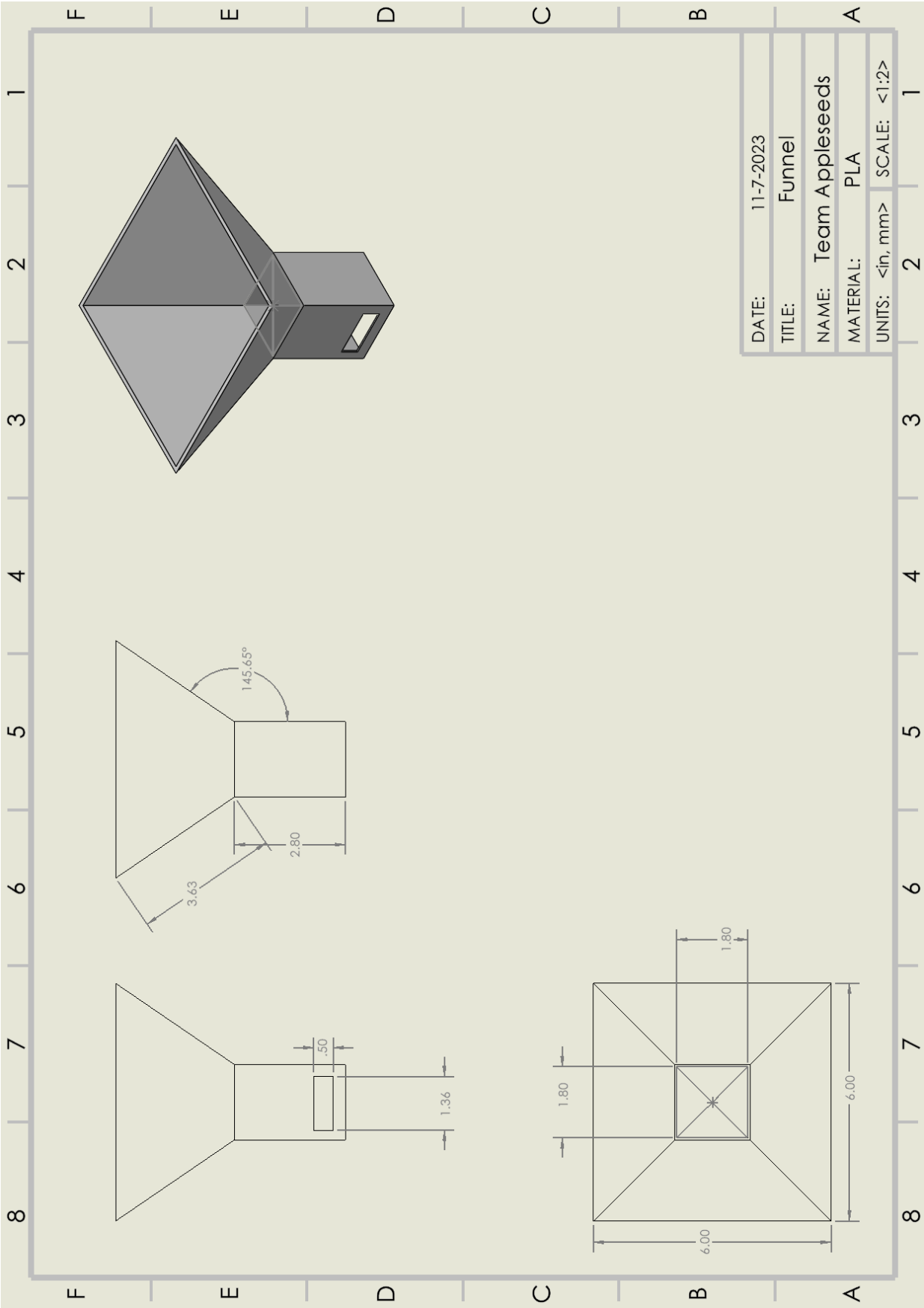


Figure 16: CAD Interior Ramps

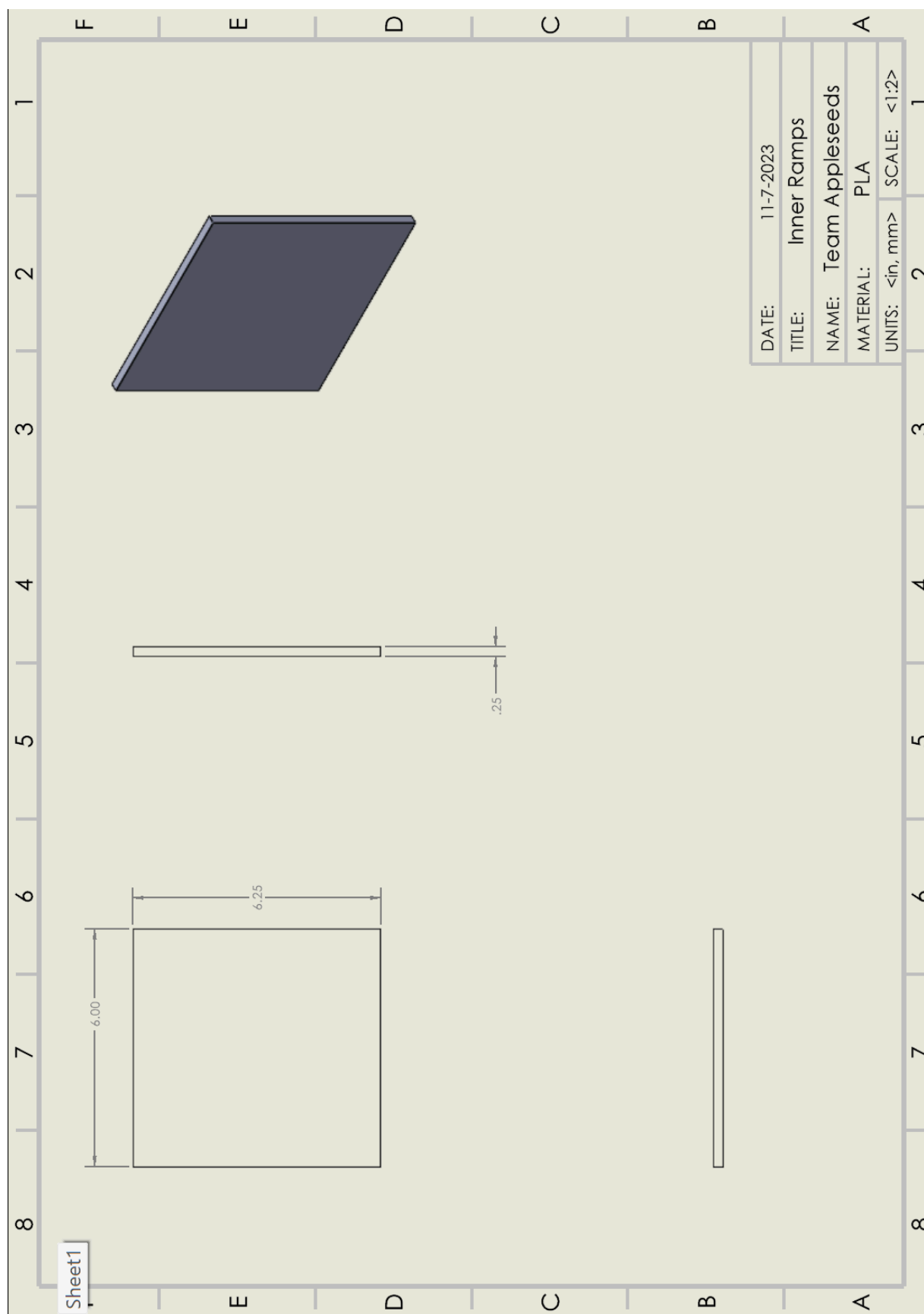


Figure 17: CAD Bottom Interior Ramp

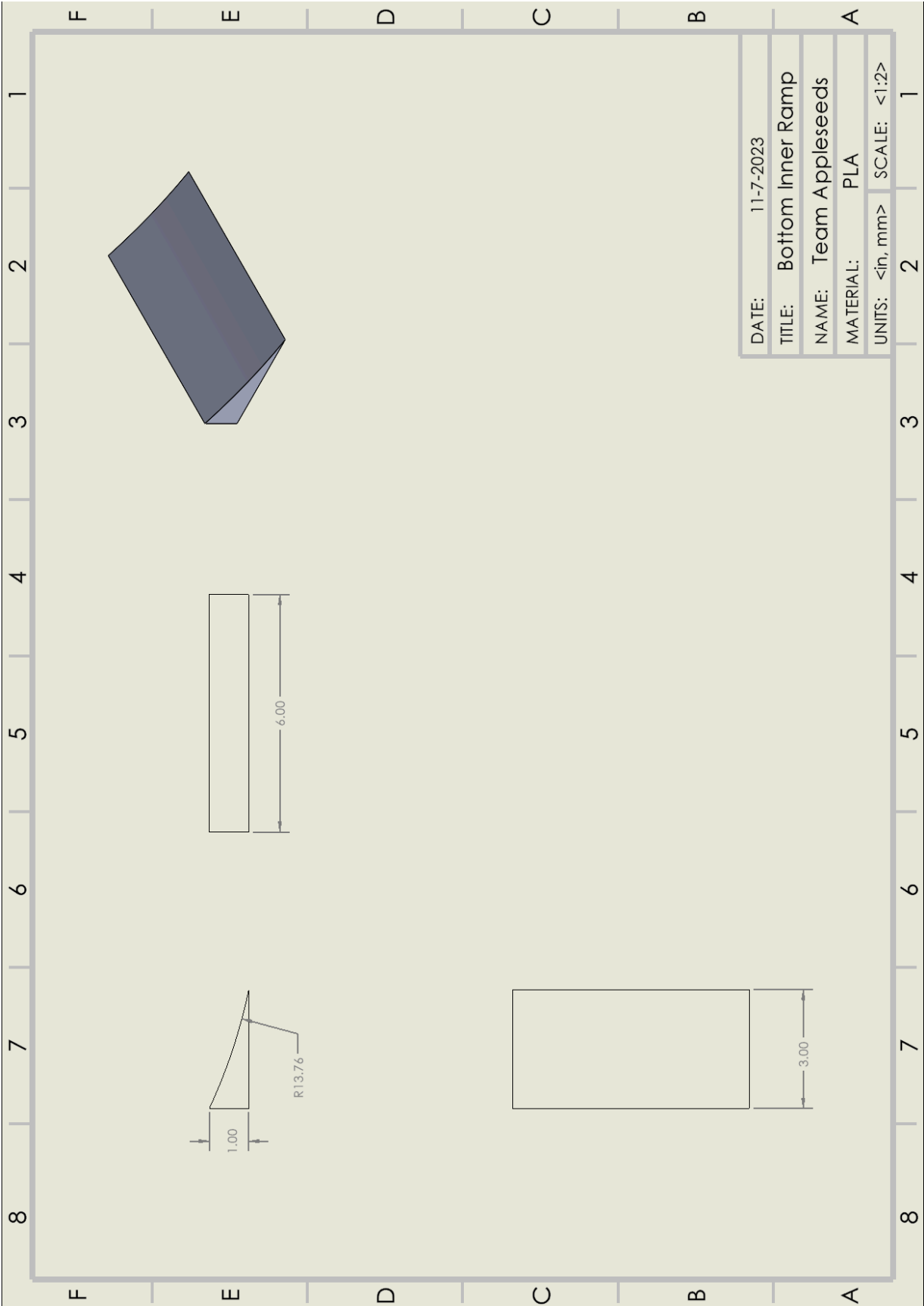


Figure 18: CAD Outer Ramp 1

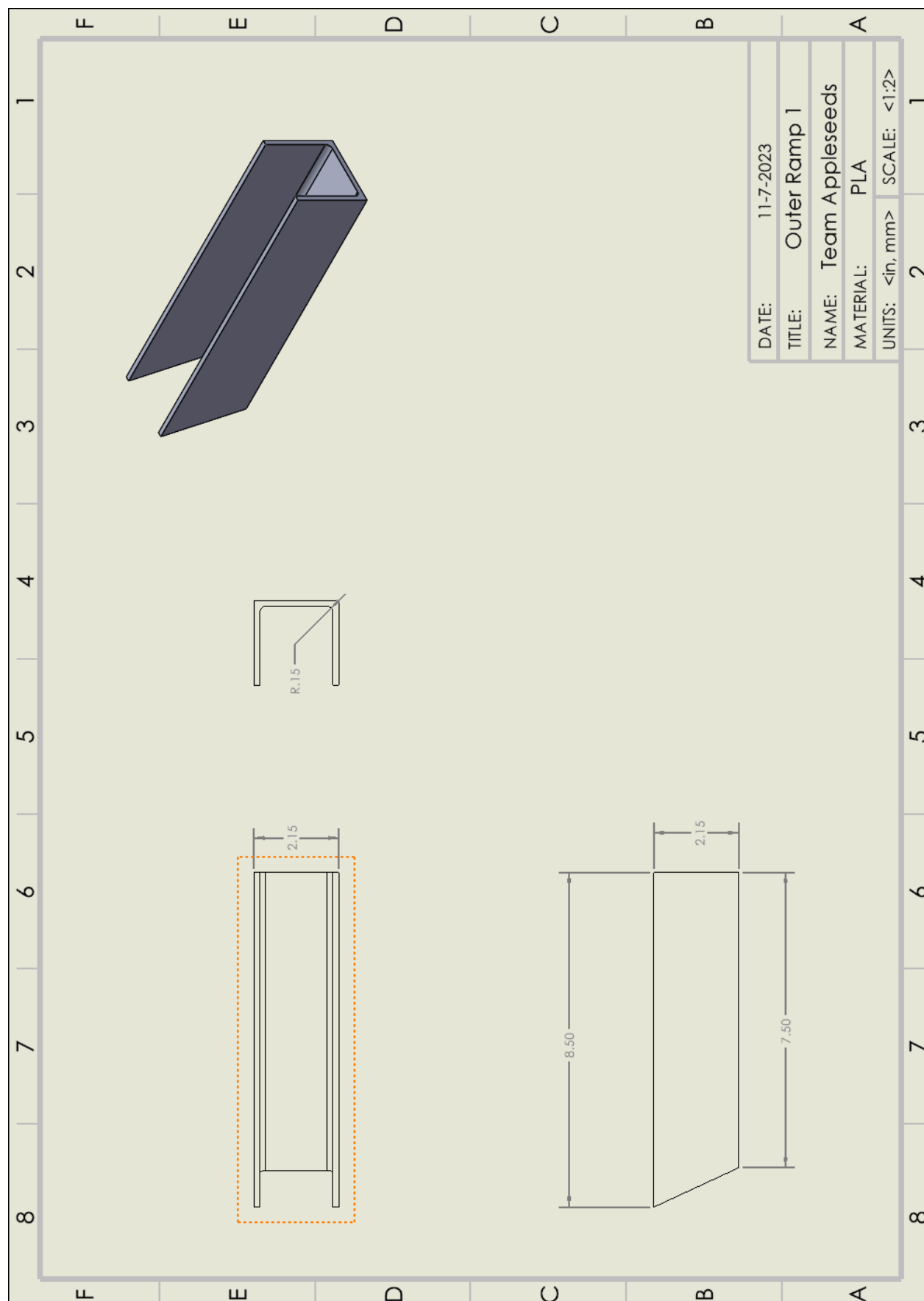
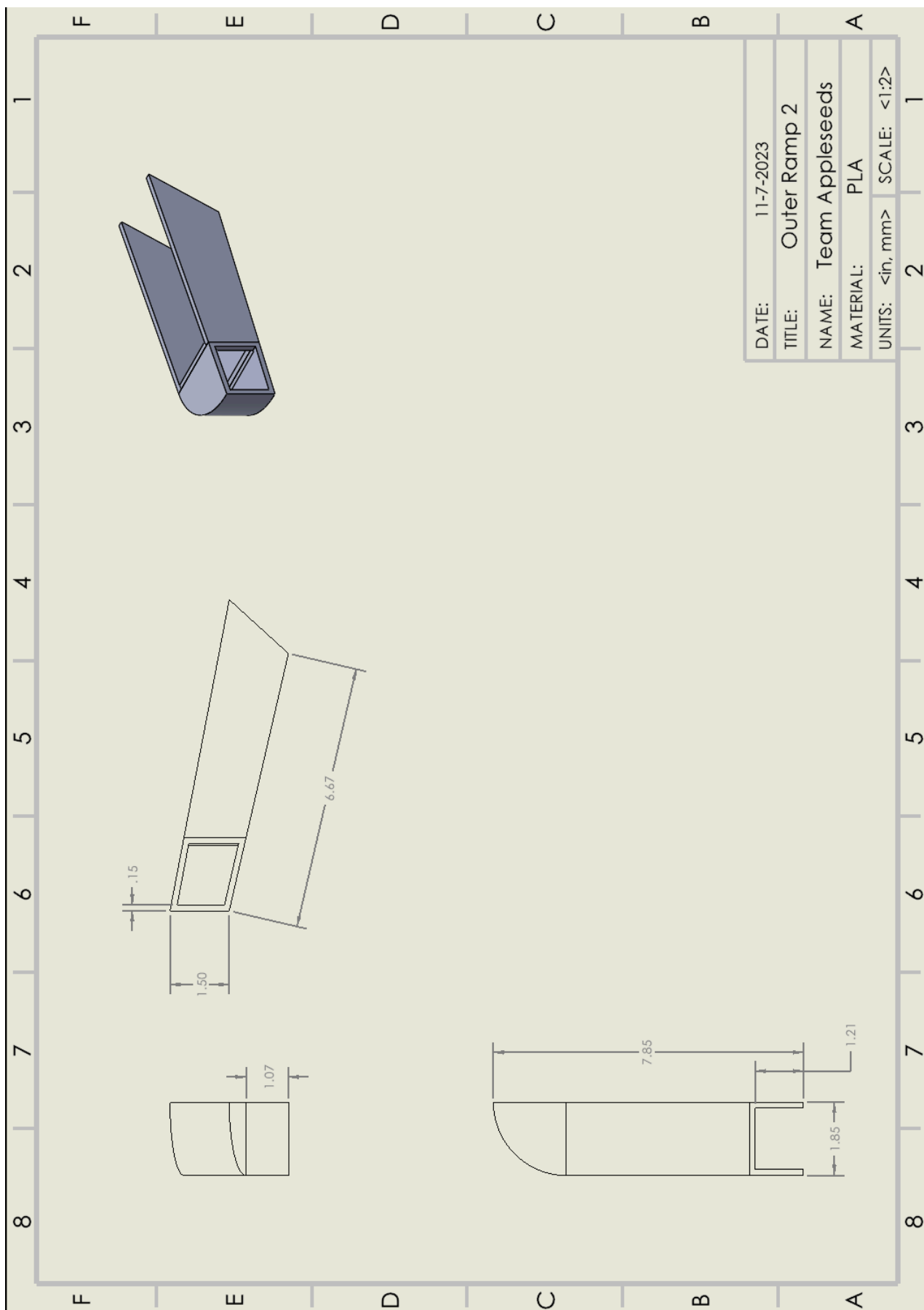


Figure 19: CAD Outer Ramp 2



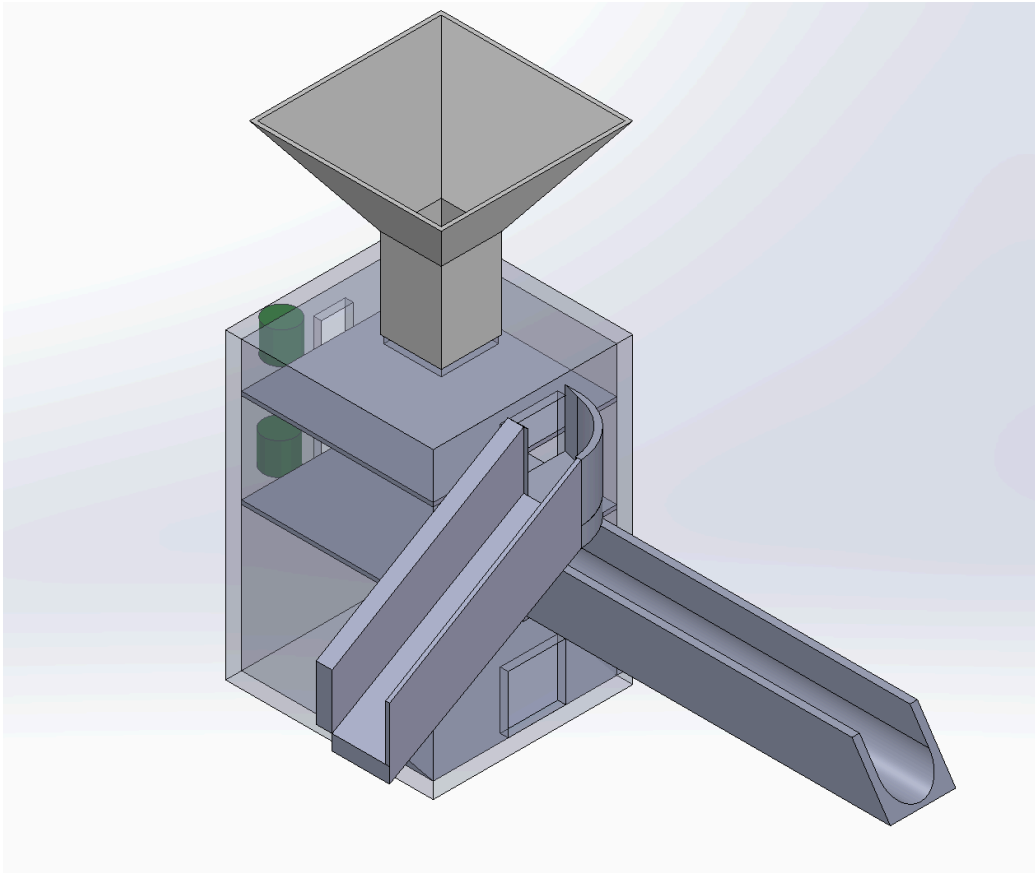


Figure 20: Finalized CAD Assembly Design

7. Bill of materials:

Part #:	Part Name:	Material:	Quantity:
1	Funnel	PLA	1
2	Gateway (connected with funnel)	PLA	1
3	Outer Ramp 1 (top)	PetG* or PLA	1
4	Outer Ramp 2 (middle)	PetG* or PLA	1
5	Inner Moving Ramp 1 (top)	PetG* or PLA	1
6	Inner Moving Ramp 2 (middle)	PetG* or PLA	2
7	Inner Ramp (bottom)	PetG* or PLA	1
8	Frame (box with cuts)	PLA or MDF	1
9	Battery	Electronic	1
10	Servo Motor	Electronic	3
11	Photoresistor	Electronic	1
12	Arduino	Electronic	1

*not available to use as a material

Table 13: Bill of Materials

Table 13 is a *Bill of Materials (BOM)*. A BOM is a table that lists the material that is used for each part, along with their part name and quantity. This systematically organizes all the parts used in the device, and is a simple way to understand what materials are needed in order to construct the device.

8. Conclusion:

Throughout the course of this project, the team has been able to draw many conclusions on how to define a problem, recognize the functions that the device needs in order to succeed, and how to execute its purpose. The team was tasked with the operation of creating a device that accurately sorts colored ping pong balls. With trial and error, the team is confident enough to commence the fabrication of this device.

To begin, the team had to create a clear understanding of what the device would resolve. With this in mind, the team began to develop a list of objectives and metrics. The objectives and metrics would aid in distinguishing what exactly the device would be capable of doing. Following that, the team created an objective tree to show the different properties needed for the device to solve the necessary problem. A function-means tree was created to expand this topic more and show a flowchart of each component needed for the device. Lastly, the team took that information from the function-means tree and inputted it into a morphological chart to eliminate possible choices in order to get to the team's top 3 concept choices. Lastly, each top sketch was thought out to show the different types of design ideas, and the team's top choice became the final concept. This final concept was then put into SolidWorks to be put together in an assembly, and each part was put into a drawing with dimensions so that it was easy to understand separately. The last step in this process was to create a bill of materials so that when the device is ready to be fabricated, it will be easy to read and manufacture in the future.

Overall, the team has learned a lot about the different aspects of the engineering design process and is learning to take those skills and keep it in mind for the next project. The number one skill that was the most helpful throughout this process was the ability to problem-solve and work as a team through any additional roadblocks. Some technical skills that were learned were how to proficiently use SolidWorks in real-life scenarios, creating flowcharts for different design aspects, and learning to use charts such as the morphological chart to help streamline the internal process of concept selection. Some challenges that were faced during this project were coming up with different design concepts that were feasible for our criteria over the course of concept selection. Going forward, the team hopes to continue this trend of problem-solving and reliable communication to complete future tasks. Using a Gantt chart throughout the project proved to be useful, as the team was able to ensure all tasks were completed in a timely manner and split up work fairly and evenly. For the next project, the team hopes to expand more on the thoughts of how to stabilize the inner ramps when attached to the servo, how to block out more light from the photoresistor, as well as how to properly fabricate the parts of their device.

9. References:

[1]“Servo Motor Basics with Arduino | Arduino Documentation.”

<https://docs.arduino.cc/learn/electronics/servo-motors>

[2] Dejan, “Arduino Color Sorter Project,” *How to Mechatronics*, Feb. 17, 2022.

<https://howtomechatronics.com/projects/arduino-color-sorter-project/>

[3]“Black and White Object sorting Robot | project,” © 2018 *ElectronicWings*.

<https://www.electronicwings.com/users/amrut/projects/22/black-and-white-object-sorting-robot>

[4] C. L. Dym. *Engineering Design: A Project-Based Introduction*. 4th ed.